



East West University

Project Report

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Table of contents: -

1. Data Description	Page 3 – 4
2. Central Tendency	Page 4 - 6
3. Distribution and shape	Page 6 - 13
4. Categorical variable analysis	Page 13 - 18
5. Relation between variables	Page 18 - 21
6. Data quality assessment	Page 21- 23
7. Key Insight	Page 24
8. Linear regression	Page 25- 27

1. **Data description:** -

The dataset was taken from Kaggle. This dataset represents the relation between student's exam scores and various factors. This will help us understand which factors have any positive or negative impact on students' exam scores.

Dataset	Observation / row	Features / Columns
	6607	20

Description of every Feature in dataset: -

Feature	Data Type	Meaning	Classification
Hours_Studied	Integer	Represents how many hours a student has studied	Numerical, Discrete
Attendance	Integer	Represents how regular a student was	Numerical, Discrete
Parental_Invlovement	String	Shows the involvement of parents	Categorical, Ordinal
Access_to_Resources	String	Shows the level of access to resources	Categorical, Ordinal
Extracurricular_Activities	String	Represents the involvement of extracurricular activities	Categorical, Binary
Sleep_Hours	Integer	Show the sleep hours of a student	Numerical, Discrete
Previous_Scores	Integer	Represents previous exam results of a student	Numerical, Discrete
Motivation_Level	String	Indicates the motivation level of a student	Categorical, Ordinal
Internet_Access	String	Shows the access of internet	Categorical, Binary
Tutoring_Sessions	Integer	Represents the sessions a student had of tutoring	Numerical, Discrete
Family_Income	String	Indicates the family income of a student	Categorical, Ordinal
Teacher_Quality	String	Indicates the level of the teacher	Categorical, Ordinal

School_Type	String	Indicates the type of school	Categorical, Binary
Peer_Influence	String	Indicates the peer influence of a student	Categorical, Ordinal
Physical_Activity	Integer	Represents the hours of physical activity a student had	Numerical, Discrete
Learning_Disability	String	Indicates if the student has any learning disability or not	Categorical, Ordinal
Parent_Education_Level	String	Represents the level of education of the parent	Categorical, Ordinal
Distance_From_Home	String	Shows the distance of school from home	Categorical, Ordinal
Gender	String	Represents the gender of the student	Categorical, Ordinal
Exam_Score	Integer	Shows the current exam marks of a student	Numerical, Discrete

2. Central tendency: -

Column	Mean	Median	Mode	Standard deviation	Variance	IQR
Hours_Studied	19.975	20	20	5.99	35.89	8
Attendance	79.97	80	67	11.55	133.35	20
Sleep_Hours	7.03	7	7	1.47	2.16	2
Previous_Scores	75.07	75	66	14.4	207.35	25
Tutoring_Sessions	1.49	1	1	1.23	1.51	1
Physical_Activity	2.97	3	3	1.03	1.06	2
Exam_Score	67.24	67	68	3.89	15.14	4

•Hours_Studied has a mean of 19.75 which means an average student studied around 19.75 hours. Median 20 means 50% of students studied less than 20 hours and 50% of students studied more than equal 20 hours. Mode 20 means most students studied for around 20 hours. Standard deviation 5.99 means most students (68%) study between 15.01 and 25.99 hours. Variance 35.89 shows that the spread is moderate and squared in 5.99. IQR 8 represents the middle 50% of students studying between 16 and 24 hours, giving us an IQR of 8 hours. This shows that the core group of students is quite consistent in their study habits.

- Attendance has a mean of 79.97, which means an average student has around 79.97 attendance. Median 80 means 50% of students studied less than 80 attendance and 50% of students studied more than equal 80 attendance. Mode 67 means most students have 67 attendances. Standard deviation 11.55 means most students (68%) have attendance between 68.45 and 91.55. Variance 133.35 shows that the spread is moderate and squared in 11.55. IQR 20 represents the middle 50% of students who have attendance between 70 and 90, which tells students are quite regular.

- Sleep_Hours has a mean of 7.03 hours, which means an average student sleeps for 7.03 hours. Median 7 means 50% of students sleep less than 7 hours and 50% of students sleep more than 7 hours. Mode 7 hours means most students sleep for 7 hours. Standard deviation of 1.47 hours means most students (68%) sleep between 5.56 hours to 8.5 hours. Variance 2.16 shows that the spread is moderate and squared in 1.47. IQR 2 represents the middle 50% of students; most students sleep from 6 to 8 hours, which tells students are getting enough sleep.

- Previous_Score has a mean of 75.07, which means an average student has got around 75.07 marks. Median 75 means 50% of students got less than 75 and 50% of students got more than 75. Mode 66 means most students got 66 marks. Standard deviation 14.4 means most students (68%) have got marks between 60.67 and 89.4. Variance 207.36 shows that the spread is moderate and squared in 14.4. IQR 25 represents the middle 50% of students who have marks between 62.5 and 87.5, which tells students to be quite good.

- Tutoring_Sessions has a mean of 1.49, which means an average student has got around 1.49 Tutoring_Sessions. Median 1 means 50% of students got less than 1.49 and 50% of students got more than 1.49. Mode 1 means most students got 1 Tutoring_Sessions. Standard deviation 1.23 means most students (68%) have got Tutoring_Sessions between 0.26 and 1.72. Variance 1.51 shows that the spread is moderate and squared in 1.23. IQR 1 represents the middle 50% of students who have marks between 0.99 and 1.98, which tells students to be active.

- Physical_Activity has a mean of 2.97 hours, which means an average student has got around 2.97 hours of Physical_Activity. Median 3 means 50% of students got less than 2 hours and 50% of students got more than 3 hours. Mode 3 means most students got 3 hours Physical_Activity. Standard deviation 1.03 means most students (68%) have got Physical_Activity between 1.94 and 4 hours. Variance 1.06 shows that the spread is moderate and squared in 1.03. IQR 2 hours

represents the middle 50% of students who have marks between 1.97 and 3.97, which tells students to be physically active.

- Exam_Score has a mean of 67.24, which means an average student has got around 67.24 marks. Median 67 means 50% of students got less than 67 and 50% of students got more than 67. Mode 68 means most students got 68 marks. Standard deviation 3.89 means most students (68%) have got marks between 63.35 and 71.13. Variance 15.1321 shows that the spread is moderate and squared in 3.89. IQR 4 represents the middle 50% of students who have marks between 65.24 and 69.24, which tells students did worse than before.

3. Distribution and Shape :-

Column	Skewness	Outlier	Heavy Tailed
Hours_Studied	Left	Yes	Right
Attendance	Left	No	No
Sleep_Hours	Right	No	No
Previous_Scores	Right	No	No
Tutoring_Sessions	Right	Yes	Right
Physical_Activity	Left	No	No
Exam_Score	Right	Yes	Right

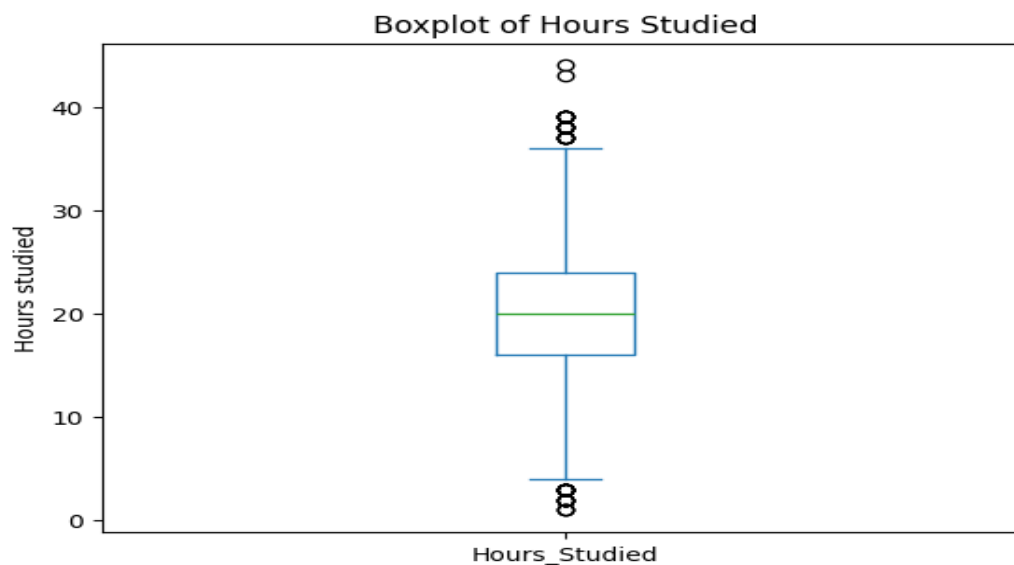


Fig 3.1 Box plot of Hours_Studied

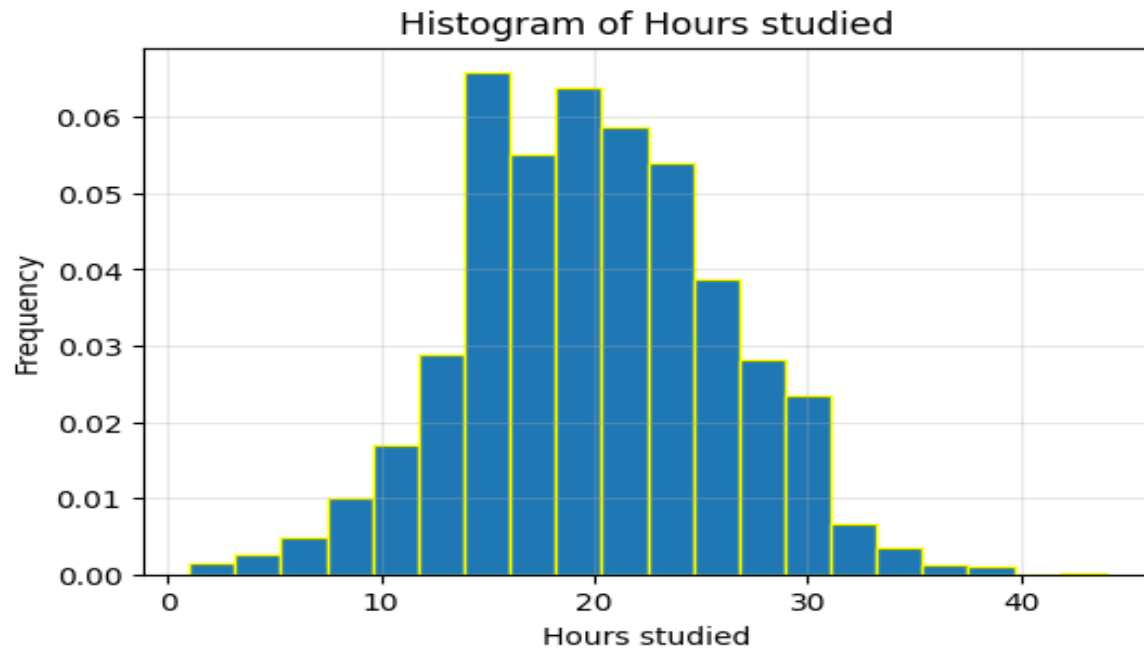


Fig 3.2 Histogram of Hours_Studied

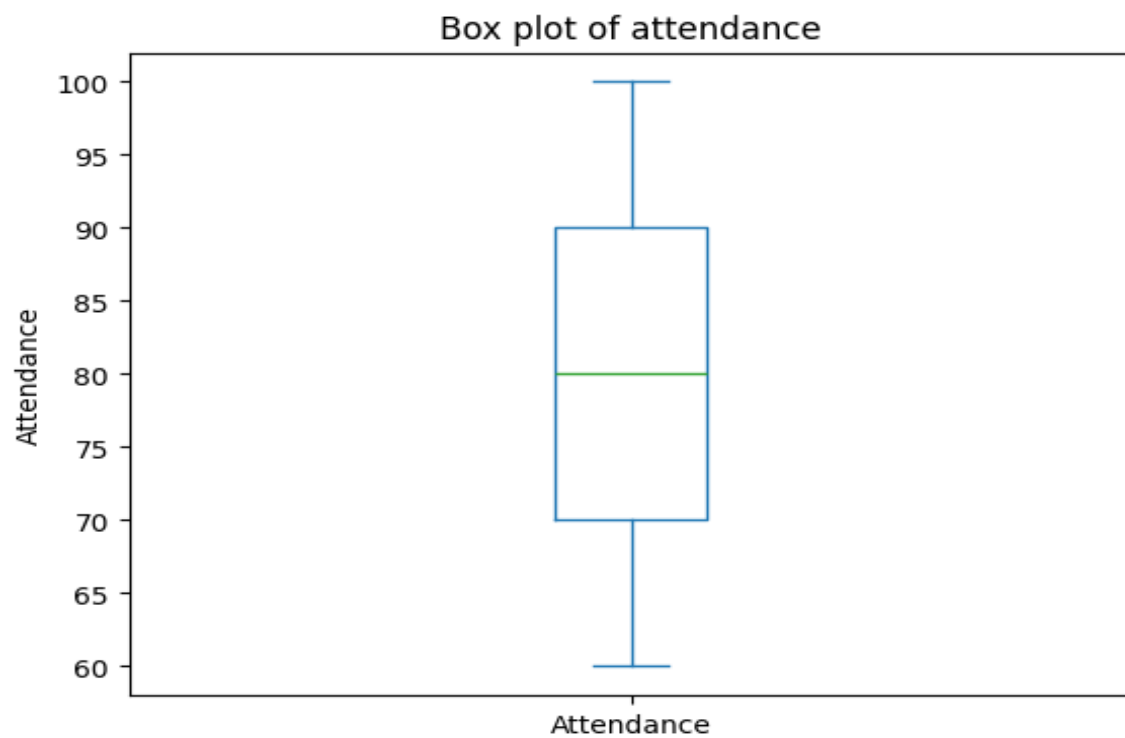


Fig 3.3 Boxplot of attendance

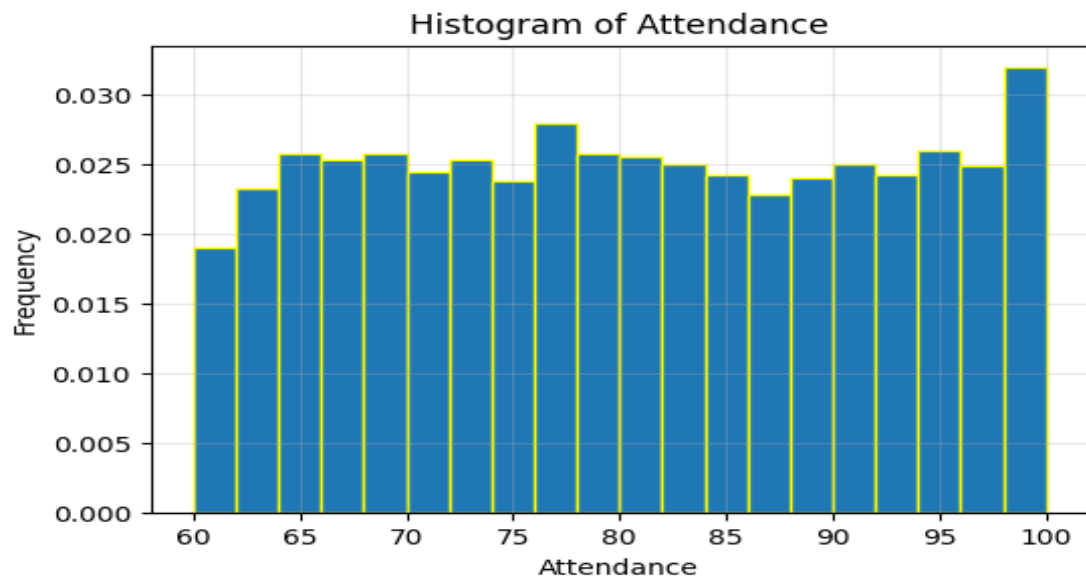


Fig 3.4 Histogram of Attendance

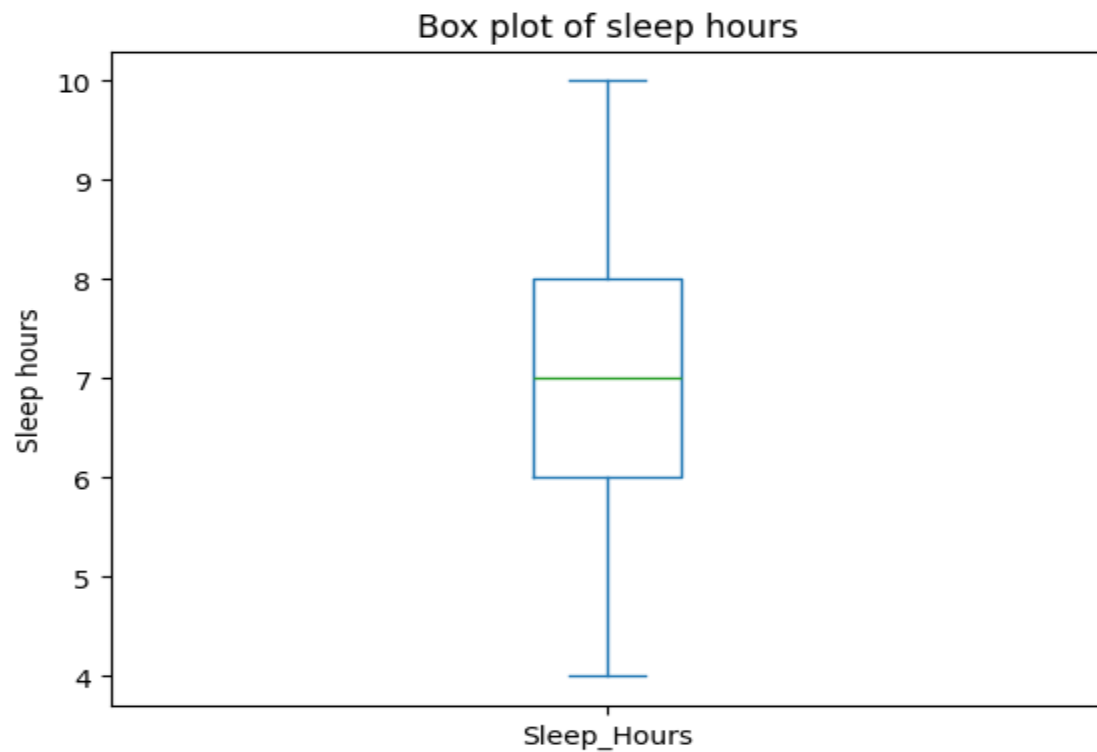


Fig 3.5 Boxplot of Sleep_Hours

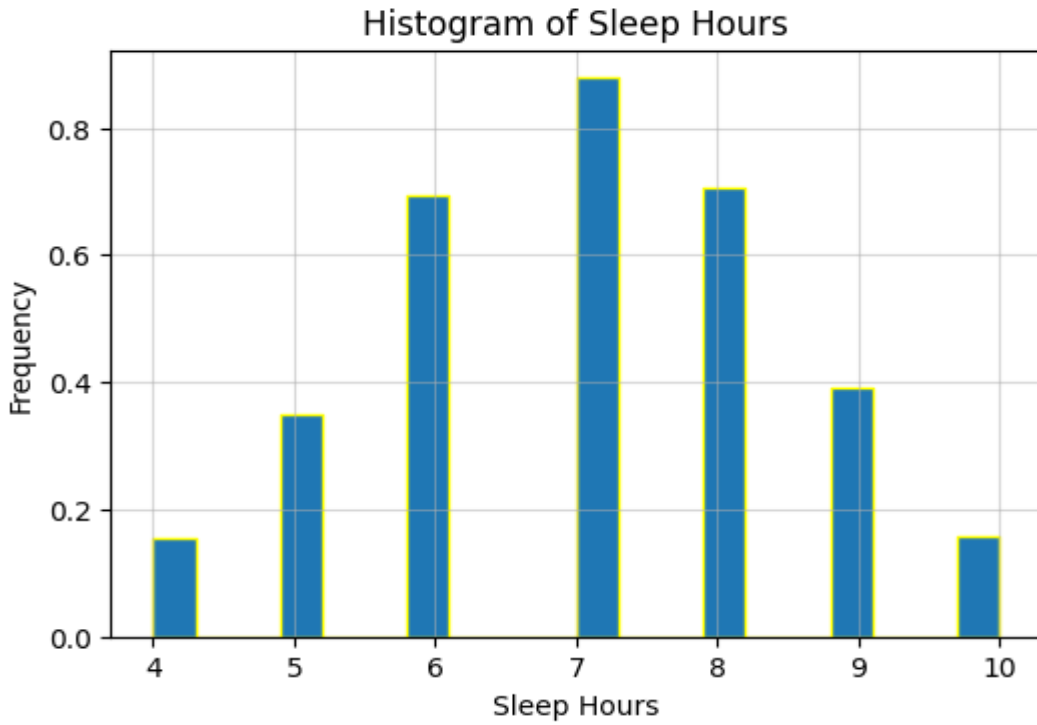


Fig 3.6 Histogram of Sleep_Hours

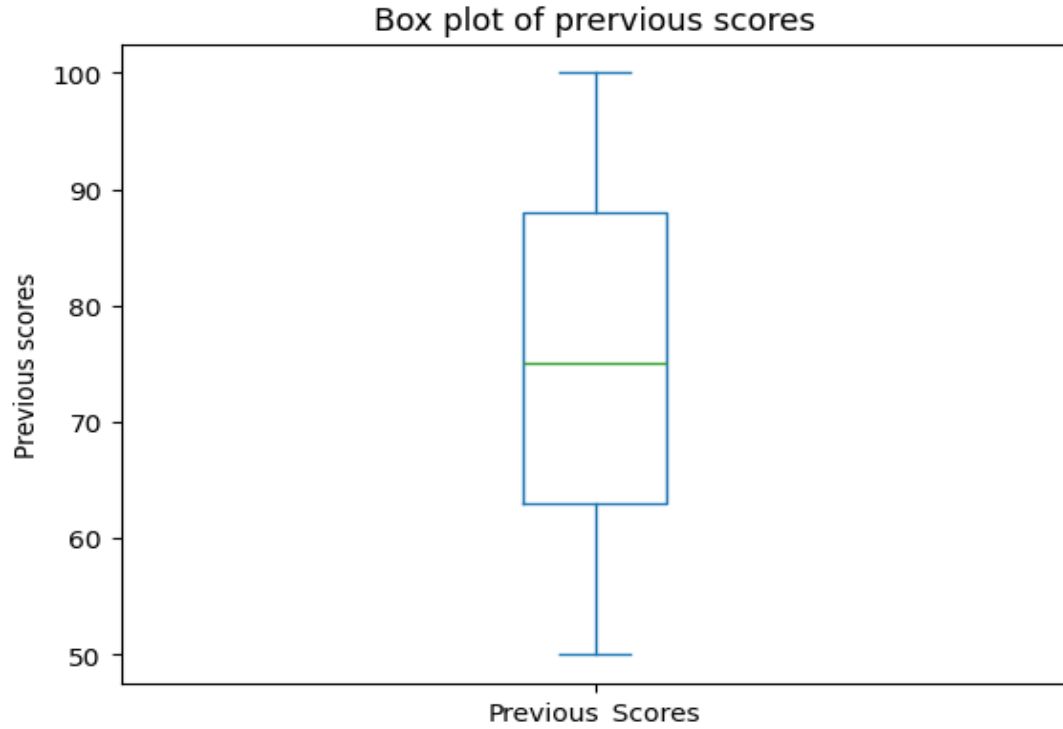


Fig 3.7 Boxplot of Previous Scores

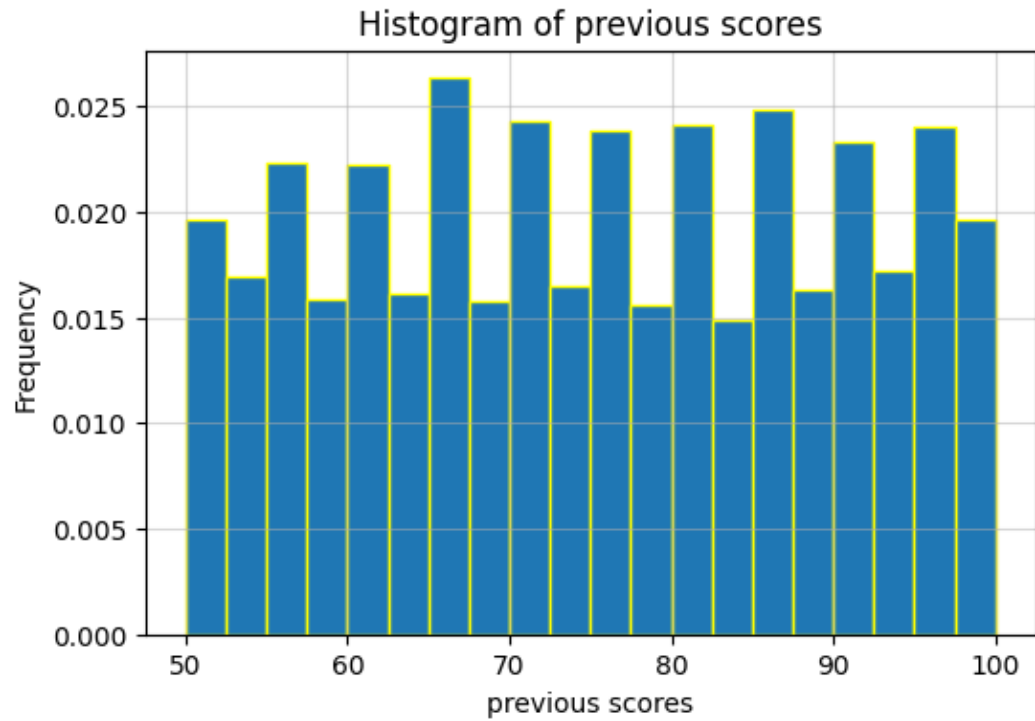


Fig 3.8 Histogram of Previous Scores

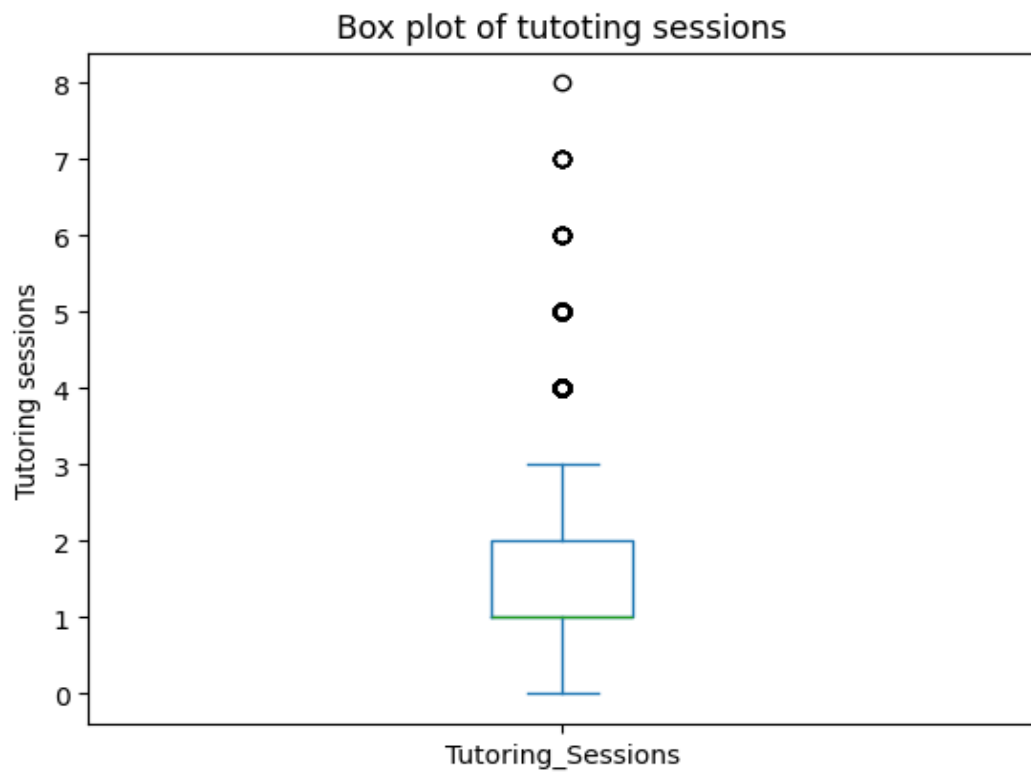


Fig 3.9 Boxplot of Tutoring_Sessions

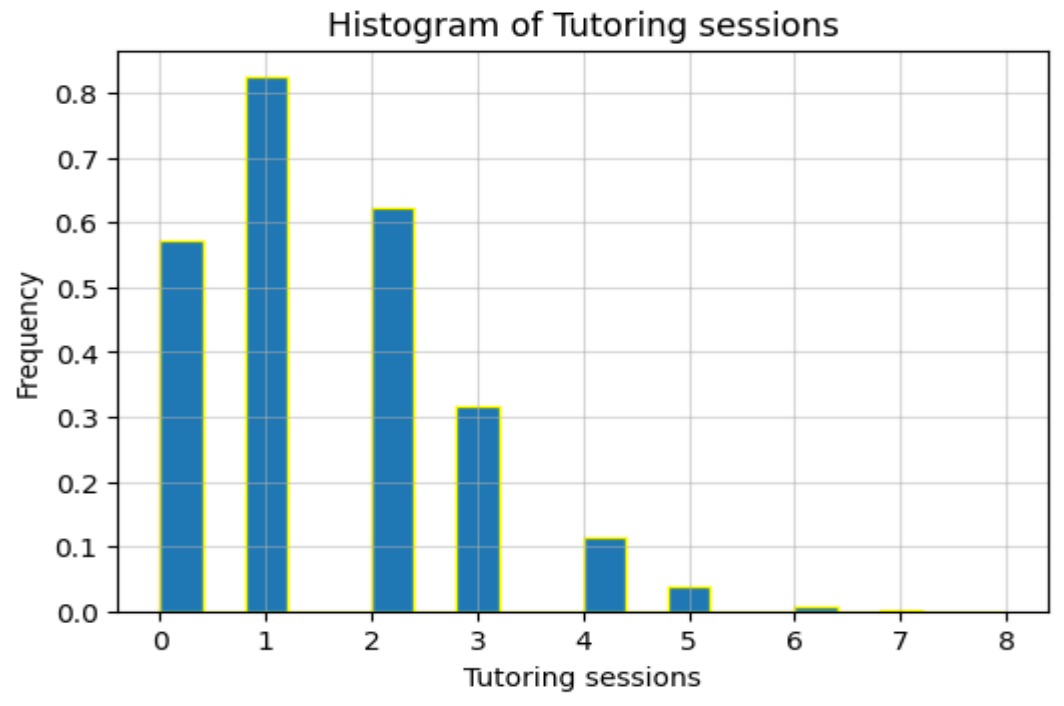


Figure 3.10 Histogram of Tutoring_Sessions

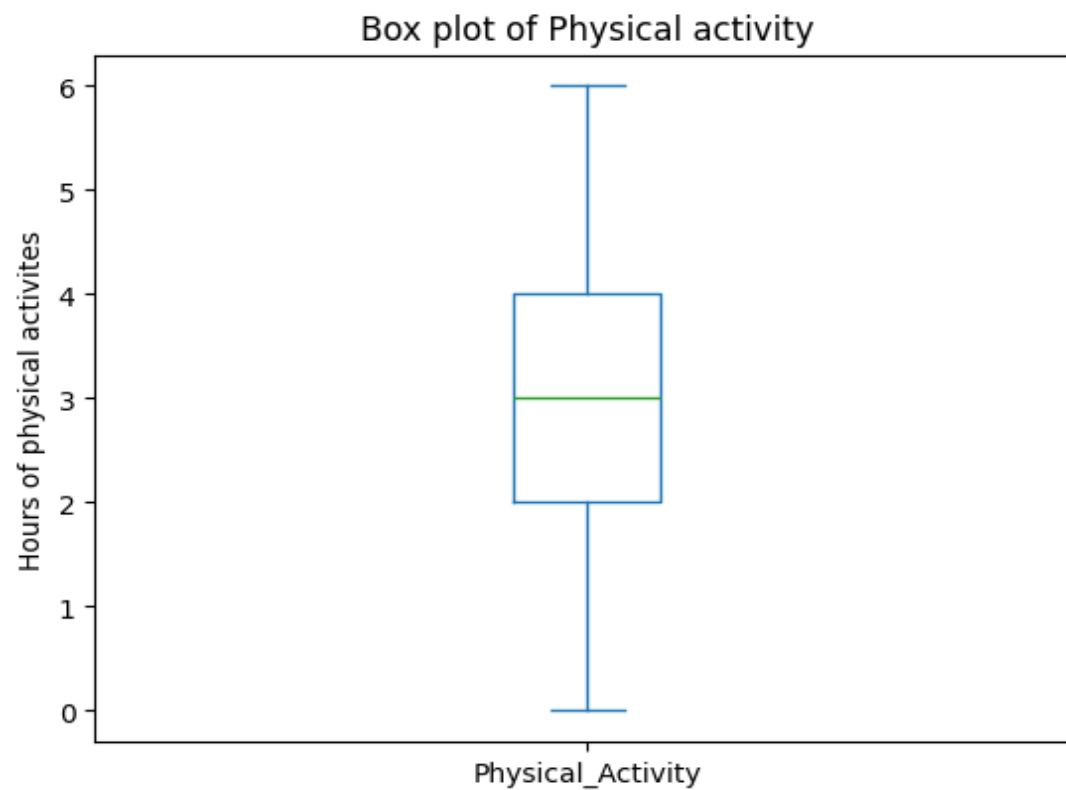


Figure 3.11 Boxplot Physical_Activity

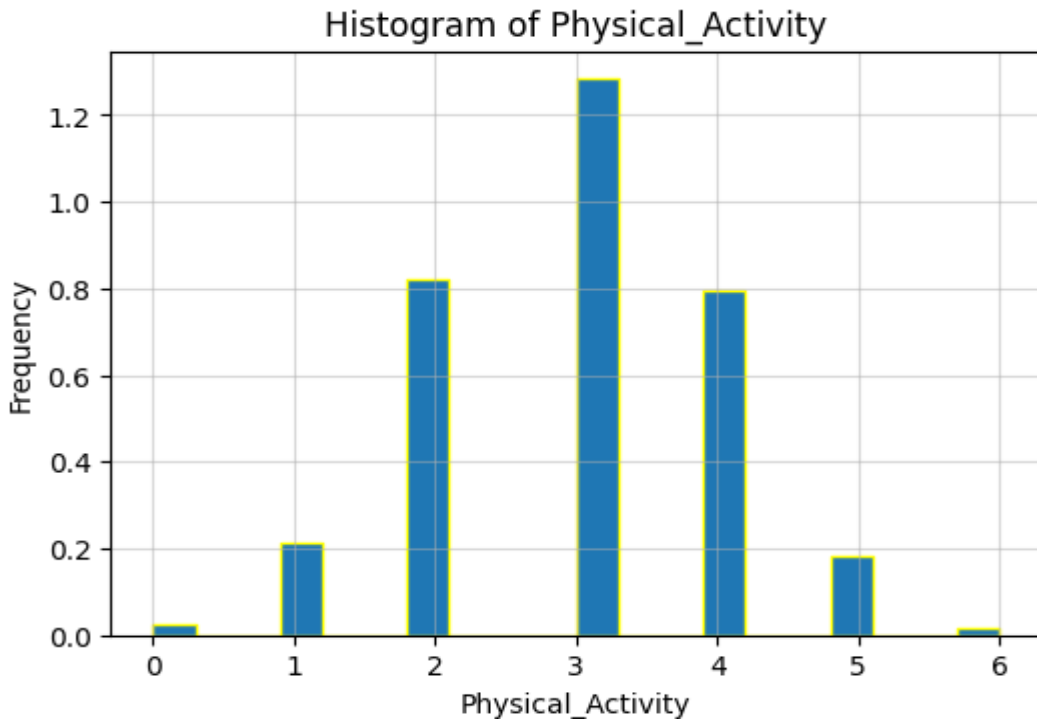


Figure 3.12 Histogram of Physical_Activity

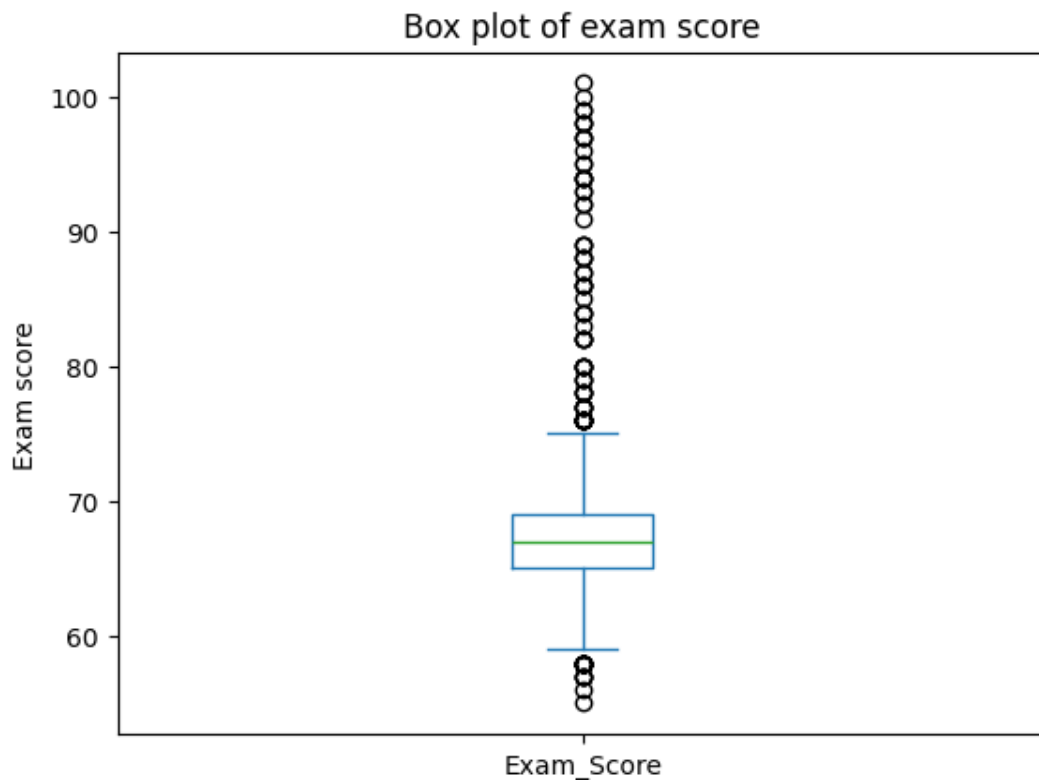


Figure 3.13 Boxplot of Exam Score

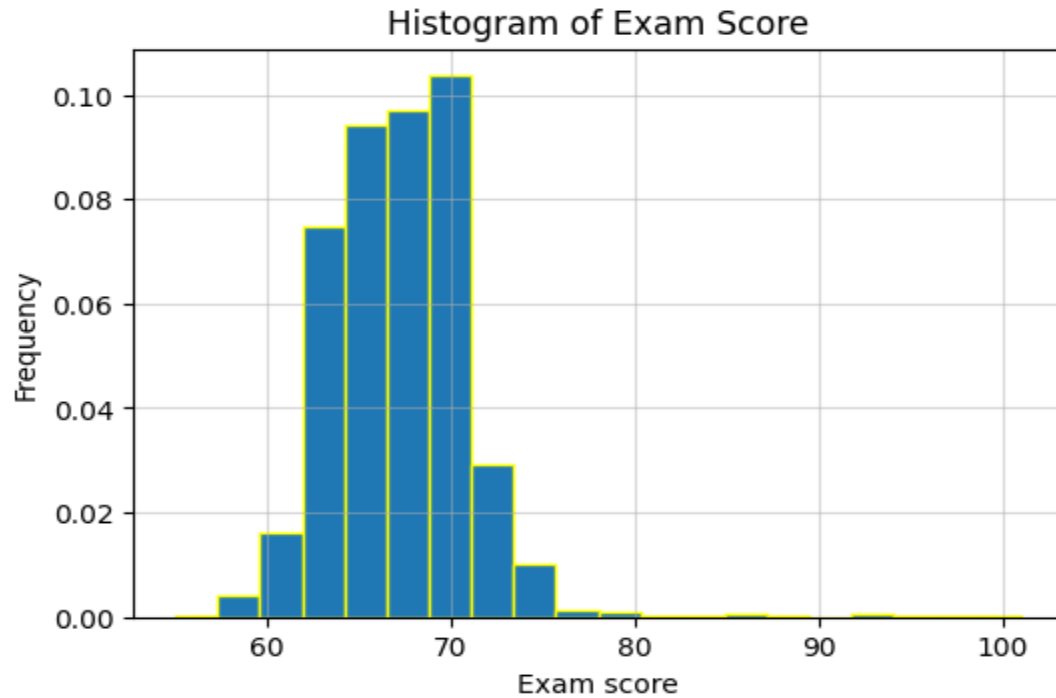


Figure 3.14 Histogram of Exam Score

4 Categorical variable analysis: -

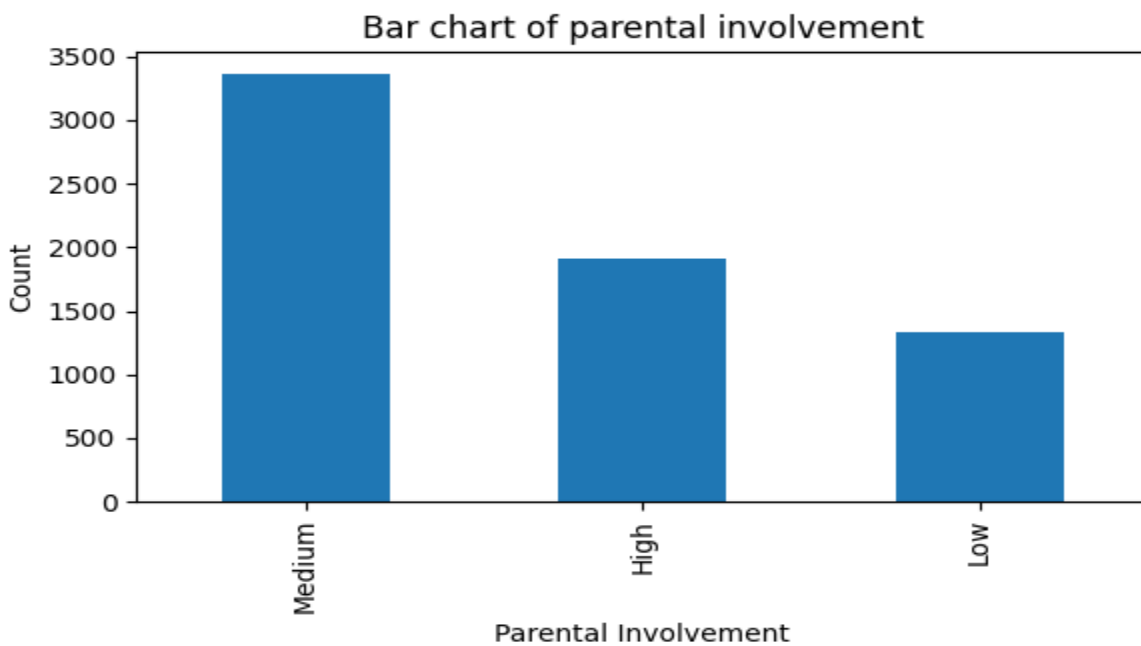


Figure 4.1 Bar chart of Parental Involvement

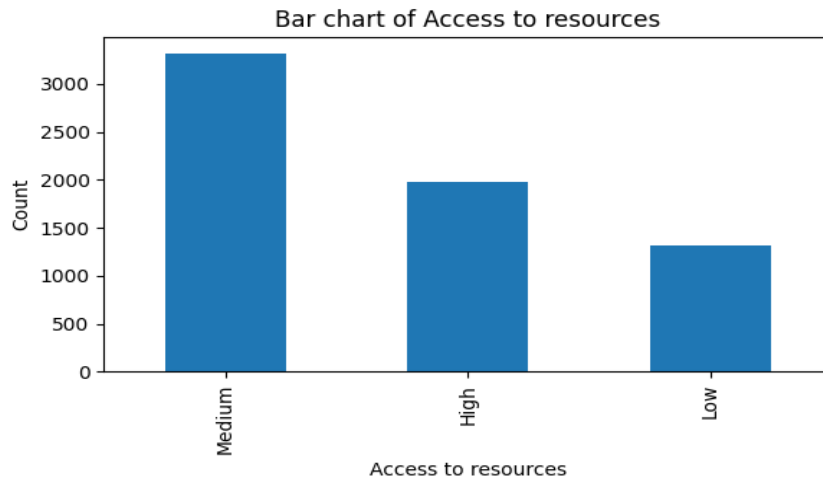


Figure 4.2 Bar Chart of Access to Resources

Pie chart of Extracurricular activities

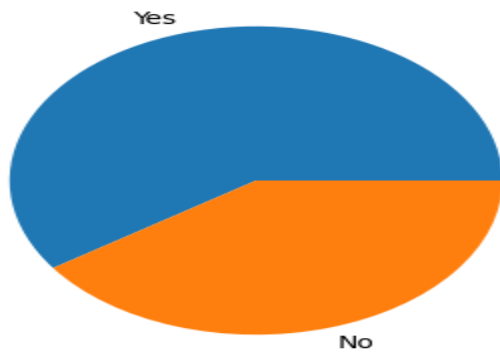


Figure 4.3 Pie Chart Of extracurricular activities

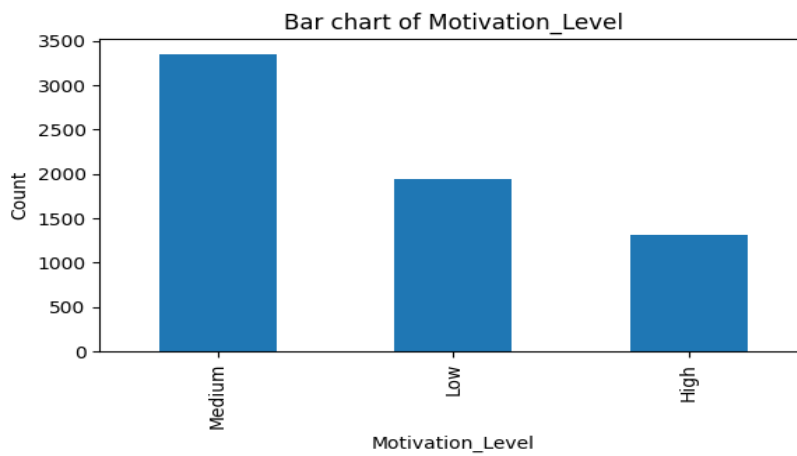


Figure 4.4 Bar chart of Motivation level

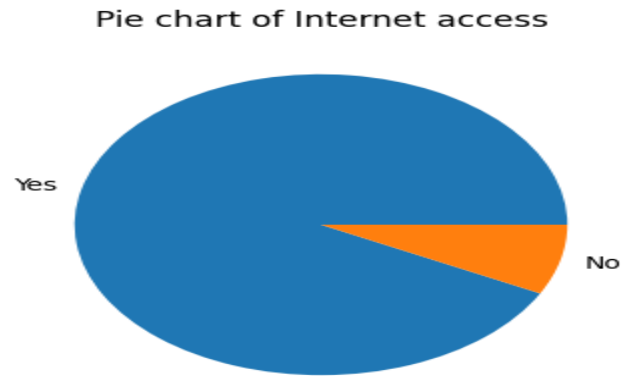


Figure 4.5 Pie Chart of Internet Access

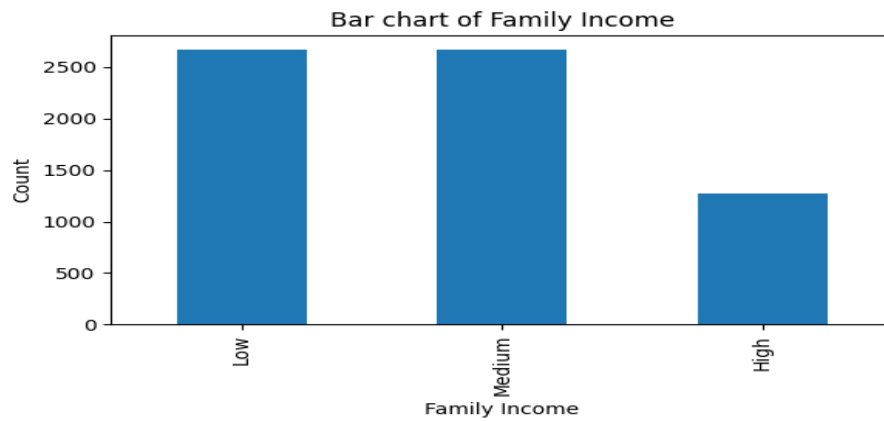


Figure 4.6 Bar chart of Family income

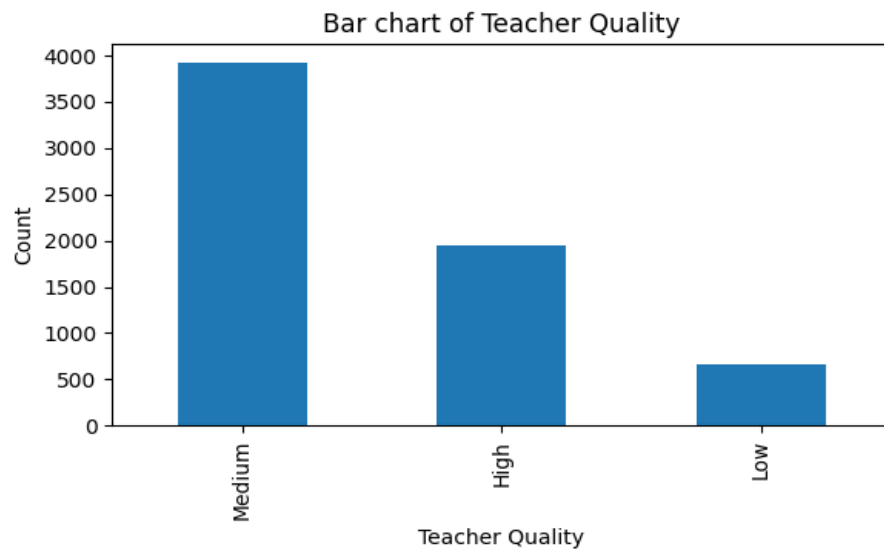


Figure 4.7 Bar chart of Teacher quality

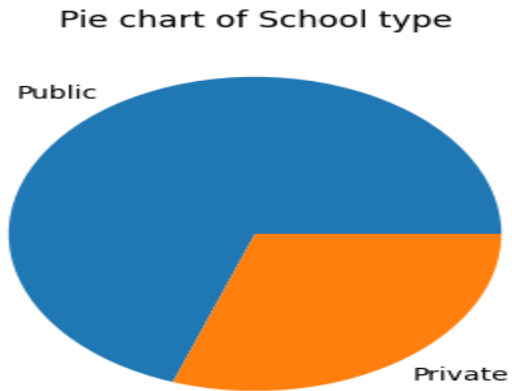


Figure 4.8 Pie Chart of School Type

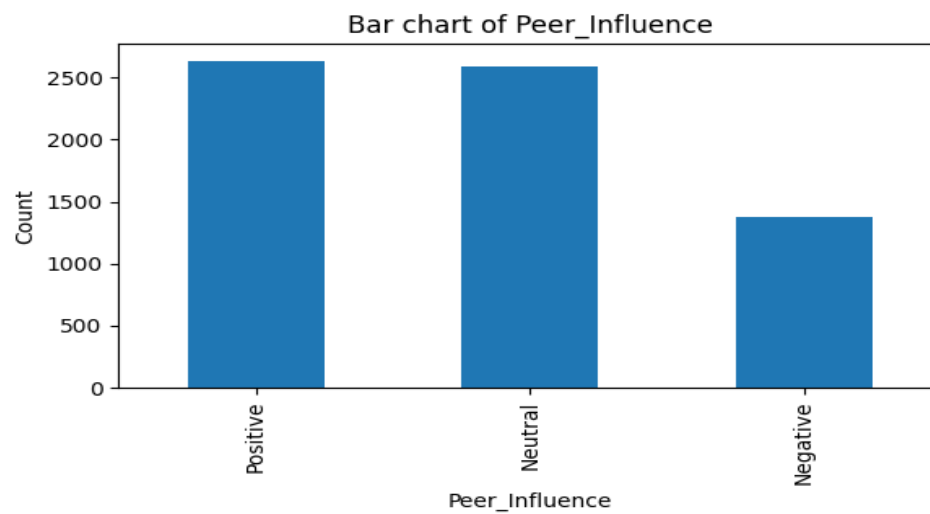


Figure 4.9 Bar Chart of Peer Influence

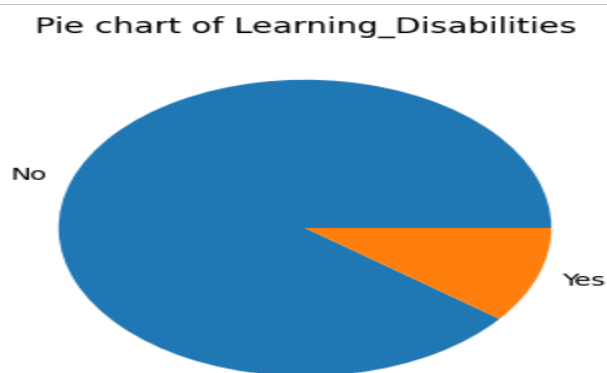


Figure 4.10 Pie chart of Learning disability

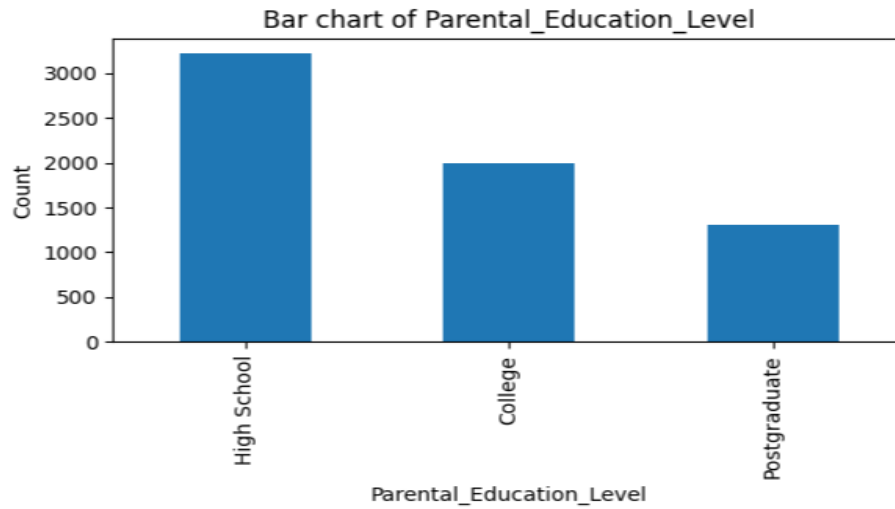


Figure 4.11 Bar chart of Parental education level

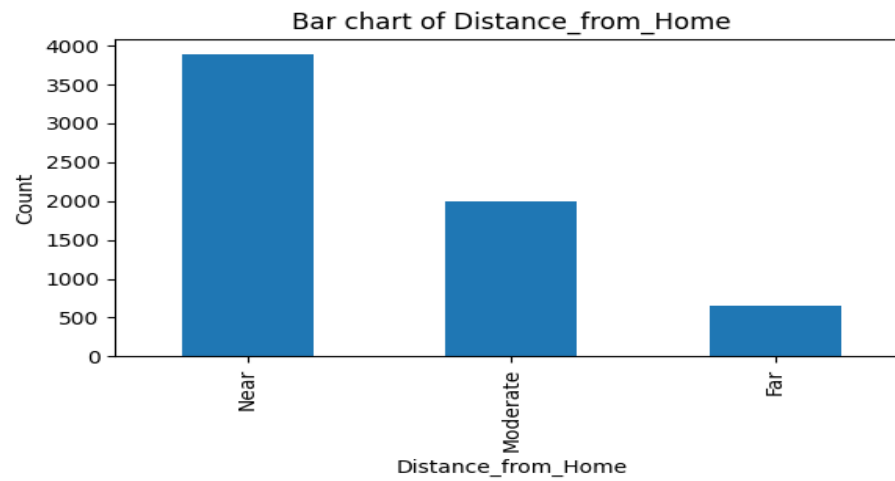


Figure 4.12 Bar chart of Distance from home

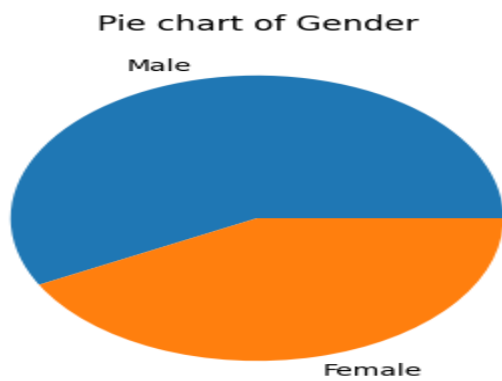


Figure 4.13 Pie chart of Gender

Column	Most appeared	Number of occurrences	Imbalance
Parental_Involvement	Medium	3362	Data occur more to the medium side
Access_to_Resources	Medium	3319	Data occur more to the medium side
Extracurricular_Activity	Yes	3938	Data occur more to Yes
Motivation_Level	Medium	3351	Data occur more to the medium side
Internet_Access	Yes	6108	Data occur more to Yes
Family_Income	Low	2672	Data occur more towards low
Teacher_Quality	Medium	3925	Data occur more to medium
School_Type	Public	4598	Data occur more towards public
Peer_Influence	Positive	2638	Data occur more towards positive
Learning_Disability	Yes	4598	Data occur more to Yes
Parental_Educational_Level	High school	3323	Data occur more towards High school
Distance_From_Home	Near	3884	Data occur more towards near
Gender	Male	3814	Data occur more towards Male

Table of Summary of all categorical columns

5.Relation between variables: -

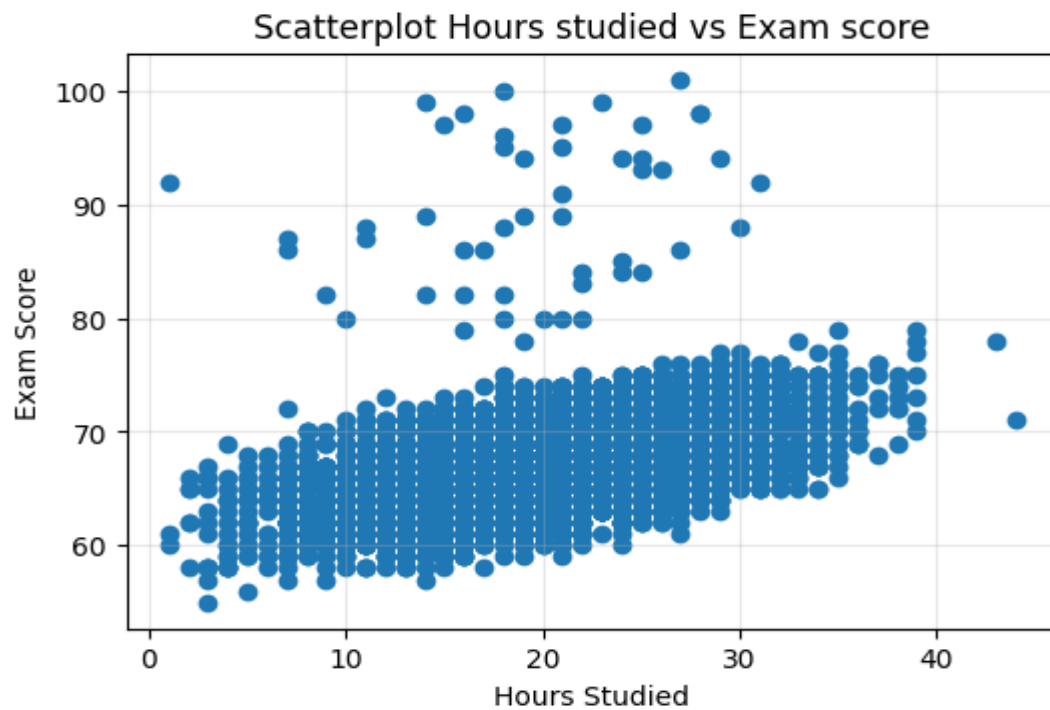


Figure 5.1 Scatterplot Between Hours_Studied and Exam_Scores

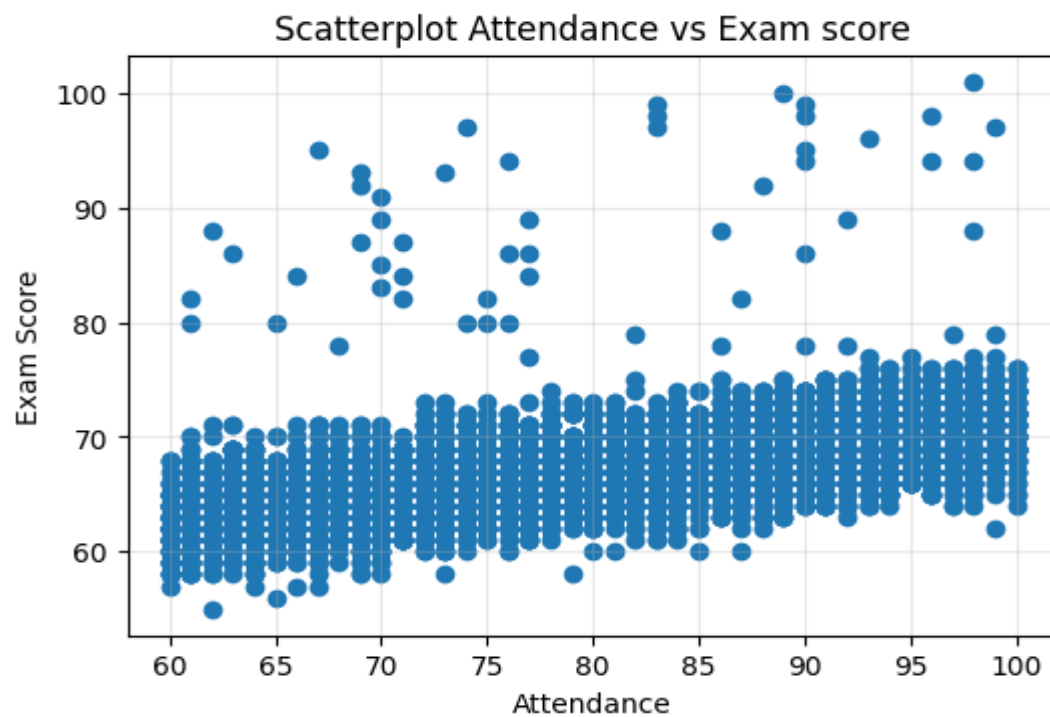


Figure 5.2 Scatterplot between Attendance and Exam_Scores

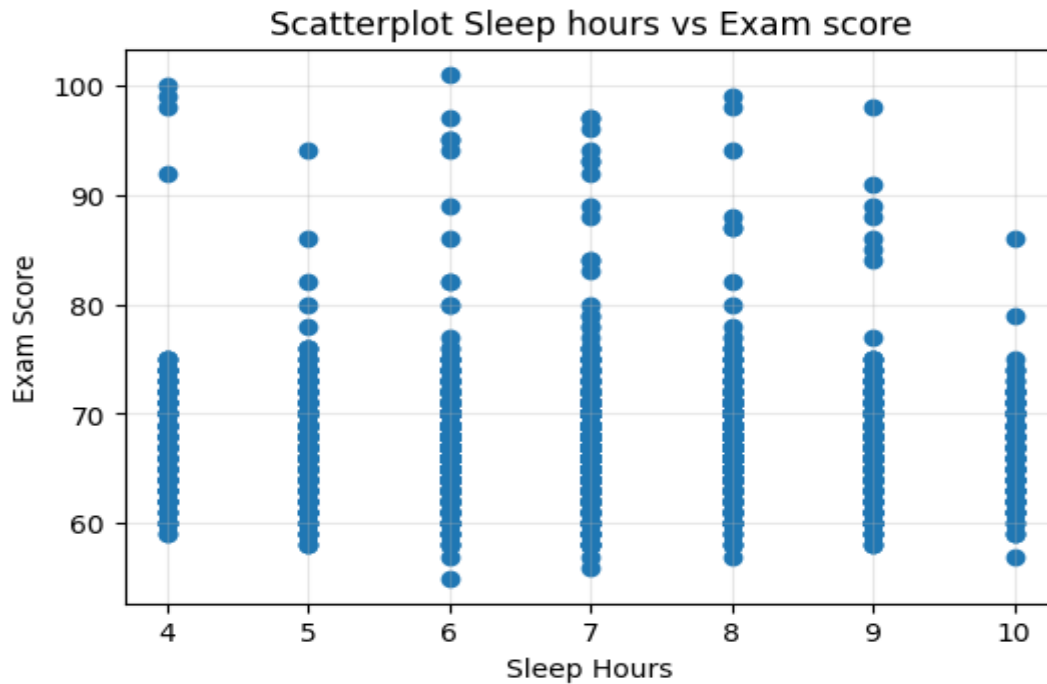


Figure 5.3 Figure between Sleep_Hours and Exam_Score

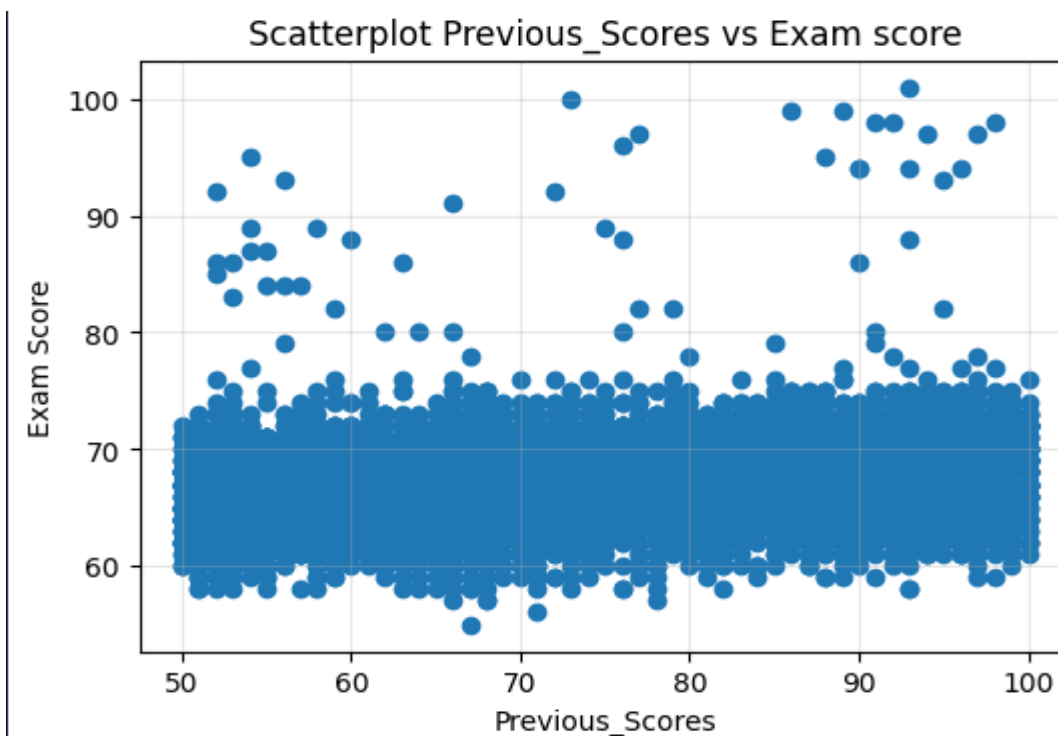


Figure 5.4 Scatterplot between Previous_Scores and Exam_Score

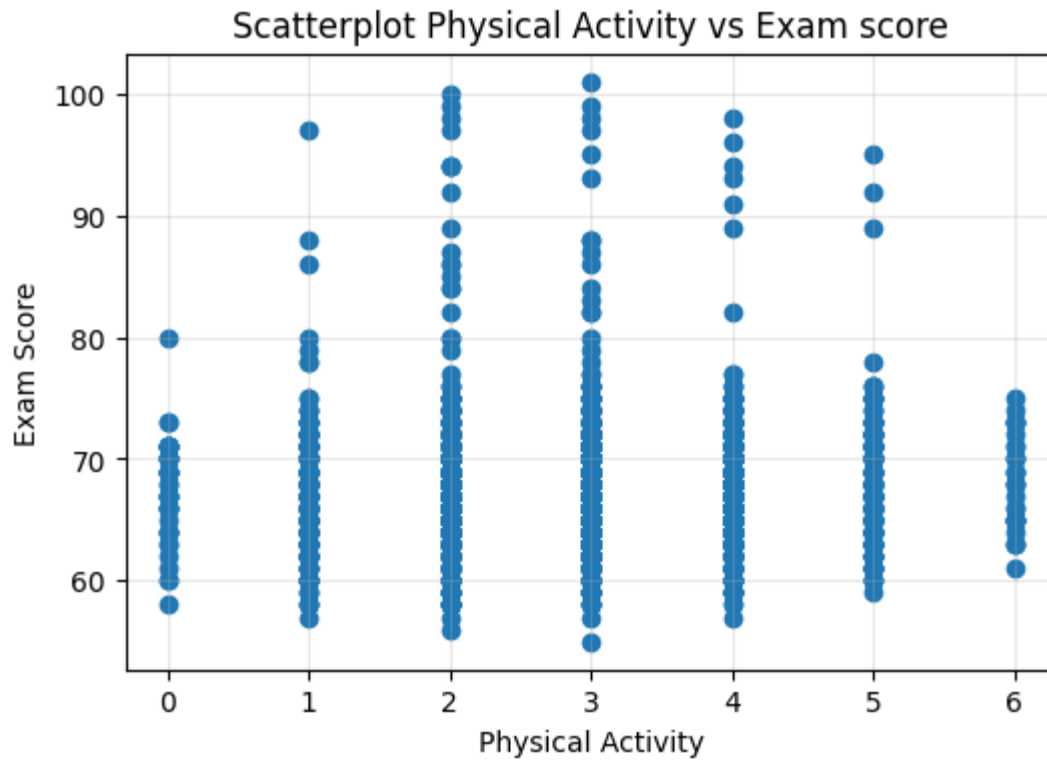


Figure 5.5 Scatterplot between Physical activity and Exam_Score

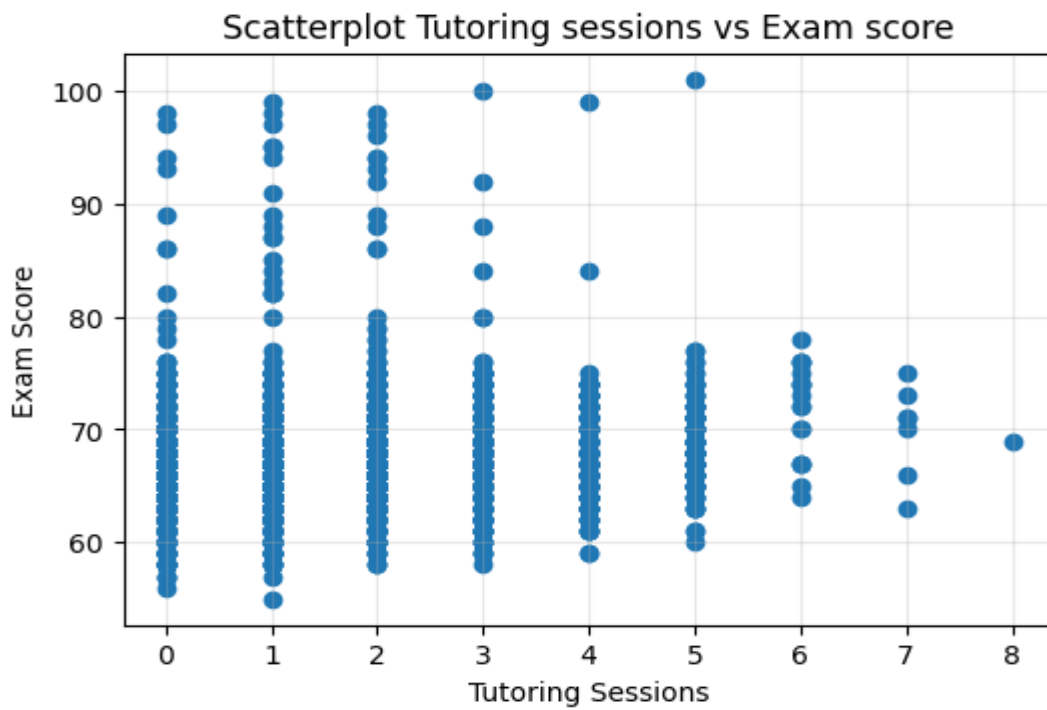


Figure 5.6 Scatterplot between Tutoring_Sessions and Exam_Score

6 Data quality assessment: -

```
print("Missing values")
print(df.isnull().sum())
print(f"TOTAL missing: {df.isnull().sum().sum()}")
```

[8] ✓ 0.0s

... Missing values

Hours_Studied	0
Attendance	0
Parental_Involvement	0
Access_to_Resources	0
Extracurricular_Activities	0
Sleep_Hours	0
Previous_Scores	0
Motivation_Level	0
Internet_Access	0
Tutoring_Sessions	0
Family_Income	0
Teacher_Quality	78
School_Type	0
Peer_Influence	0
Physical_Activity	0
Learning_Disabilities	0
Parental_Education_Level	90
Distance_from_Home	67
Gender	0
Exam_Score	0

dtype: int64
TOTAL missing: 235

Figure 6.1 List of all missing value

```
print("Duplicates")
print(f"Duplicate rows: {df.duplicated().sum()}")
```

[9] ✓ 0.0s

... Duplicates
Duplicate rows: 0

Figure 6.2 List of all Duplicate rows

```
print("\n=== OUTLIERS (All Numeric Columns) ===")
numeric_cols = df.select_dtypes(include=[np.number]).columns
Q1 = df[numeric_cols].quantile(0.25)
Q3 = df[numeric_cols].quantile(0.75)
IQR = Q3 - Q1
outliers = ((df[numeric_cols] < (Q1 - 1.5 * IQR)) | (df[numeric_cols] > (Q3 + 1.5 * IQR))).sum()
print(outliers)
```

[10] ✓ 0.0s

...

```
=== OUTLIERS (All Numeric Columns) ===
Hours_Studied      43
Attendance          0
Sleep_Hours        0
Previous_Scores    0
Tutoring_Sessions  430
Physical_Activity   0
Exam_Score         104
dtype: int64
```

Figure 6.3 List of all outliers

Missing values, duplicate rows, and outliers can easily manipulate data. They can also result in changing the distribution and skewness which can give unreliable and irrelevant results which can further change the outcome and can make the data and experiment leading to errors and rejections.

For error handling, Python has some great in-built features and libraries which can solve the following problems. Which makes the data set more reliable and makes the experiment more accurate. Also, for data selection we need to follow measures and use certified sources only which will minimize the chances of error and data Manipulation.

7 **Key Insights: -**

In this dataset, we saw factors that have an impact on students exam scores. We divided the 20 features into 2 parts in numeric and categorical. Numeric columns had all discrete values. On the other hand, categorical were split into Ordinal and Binary. In numerical features we saw the distributions were likely symmetric, if not then it had slight movements towards the left or right side. But, in categorical columns without a few exceptions, it has shifted significantly towards one side quite heavily. Also, numeric columns had a few outliers and abnormal values in Hours_Studied, Tutoring_Sessions and Exam_Score. For Numerical features we used box plots and Histogram for visual representation. Where for ordinal categorical features we used bar charts and for binary features we used pie charts. In final observation, we noticed Previous_Scores were better than the current Exam_Score and students performed better.

Column	Significance
Hours_Studied	Directly related, students that had mean or similar study hours performed far better than the other students.
Previous_Scores	Much important, as observations indicate that those students who did well in the first exam also did similarly in the next exam.
Attendance	Very impactful, as the higher attendance shows students did well than the other students.
Exam_Score	This is the main factor of this dataset, and this is most important for future modeling, sampling, and observations.
Tutoring_Session	Inversely related, shows that after the mean the more sessions students had performed worse than others.

Table of columns that had the most impact

8 Linear regressions: -

```
AT = A_train.T .....# A transpose
ATA = AT @ A_train .....# A'A
ATA_inv = np.linalg.inv(ATA) .....# (A'A)^-1
ATy = AT @ y_train .....# A'y

beta = ATA_inv @ ATy .....# final coefficients

print('Intercept (beta_0): .....', round(beta[0], 2))
print('Coefficient for Hours_Studied: .....', round(beta[1], 2))
print(f"\nEquation: Exam_Score = {beta[0]:.2f} + {beta[1]:.2f} x Hours_Studied")
✓ 0.0s

Intercept (beta_0):          61.34
Coefficient for Hours_Studied:  0.29

Equation: Exam_Score = 61.34 + 0.29 x Hours_Studied
```

Figure 8.1 Equation corresponding to our data set.

Intercept C	Coefficient m	Dependent variable
61.34	0.29	Hours_Studied

```
y_pred_train = A_train @ beta

# Look at first 5 actual vs predicted prices
for i in range(5):
    print(f'Actual: {y_train[i]:.0f}    Predicted: {y_pred_train[i]:.0f}')
✓ 0.0s

Actual: 65    Predicted: 67
Actual: 65    Predicted: 68
Actual: 71    Predicted: 67
Actual: 64    Predicted: 65
Actual: 66    Predicted: 67
```

Figure 8.2 Predicting data

```
y_pred_test = A_test @ beta

# MSE
errors      = y_test - y_pred_test
squared     = errors ** 2
mse         = squared.mean()
rmse        = mse ** 0.5

print('Test MSE: ', round(mse, 2))
print('Test RMSE:', round(rmse, 2), '(marks)')
```

1] ✓ 0.0s

Test MSE: 7.63
Test RMSE: 2.76 (marks)

Figure 8.3 Total error calculation

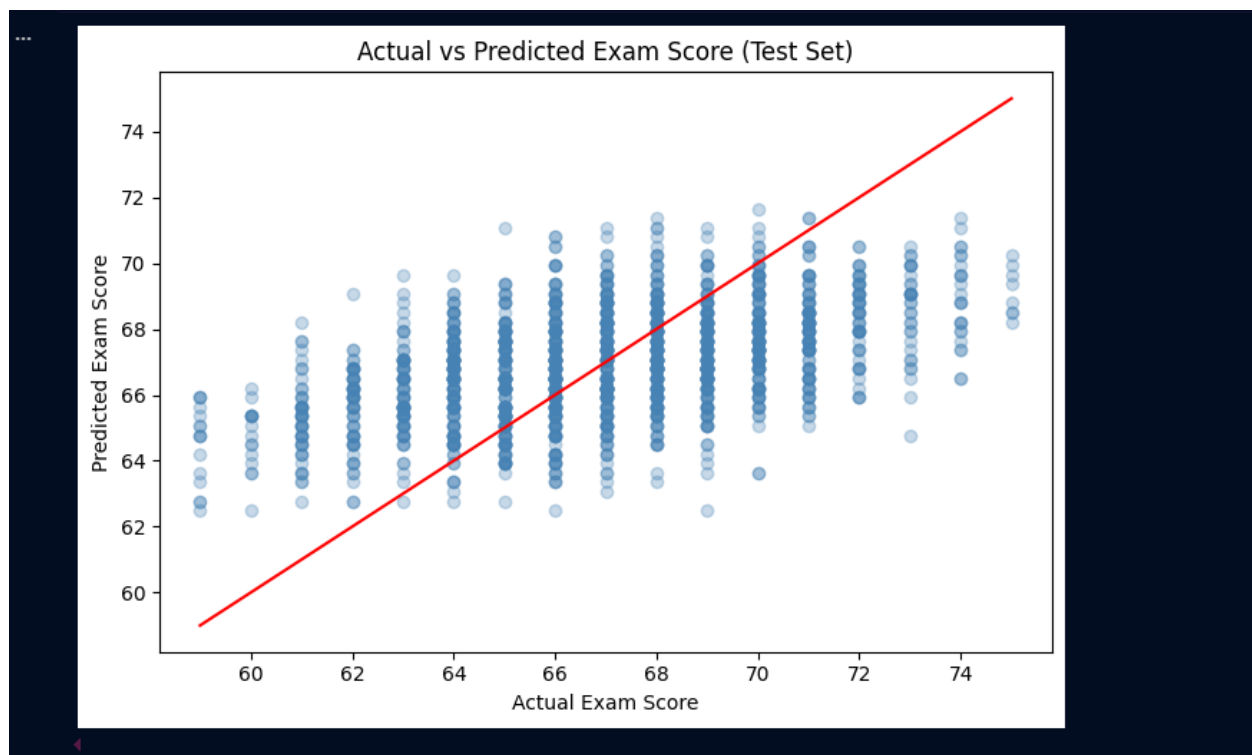


Figure 8.4 Visualization of the line

```
▷ ▾  
# categorical columns we want to include  
cat_cols = ['School_Type', 'Parental_Involvement', 'Access_to_Resources']  
  
# one-hot encode  
train_dummies = pd.get_dummies(train[cat_cols], drop_first=True)  
test_dummies = pd.get_dummies(test[cat_cols], drop_first=True)  
  
# combine numerical features with the new dummy columns  
X_train_full = np.hstack([X_train, train_dummies.values.astype(float)])  
X_test_full = np.hstack([X_test, test_dummies.values.astype(float)])  
  
# add bias column  
A_train_full = np.hstack([np.ones((X_train_full.shape[0], 1)), X_train_full])  
A_test_full = np.hstack([np.ones((X_test_full.shape[0], 1)), X_test_full])  
  
# normal equation  
beta_full = np.linalg.inv(A_train_full.T @ A_train_full) @ A_train_full.T @ y_train  
  
# predict and evaluate  
y_pred_full = A_test_full @ beta_full  
mse_full = ((y_test - y_pred_full) ** 2).mean()  
rmse_full = mse_full ** 0.5  
  
print('With categorical features - Test RMSE:', round(rmse_full, 2), '(marks)')  
print('Without categorical features - Test RMSE:', round(rmse, 2), '(marks)')  
[53] ✓ 0.0s  
... With categorical features - Test RMSE: 2.65 (marks)  
Without categorical features - Test RMSE: 2.76 (marks)
```

Figure 8.5 Utilizing categorical data